

Philosophy of Space and Time

Lecture 3: *Epistemology of Geometry*

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Today's Plan

- 1 Recap: Non-Euclidean Geometries
- 2 Epistemology of Geometry
- 3 Poincaré's Conventionalism
- 4 Discussion and Looking Ahead

Where We Left Off

Last time: three consistent geometries, distinguished by the parallel postulate.

	Euclidean	Elliptic	Hyperbolic
Parallels	Exactly one	None	Many
$\triangle$ angles	$= 180$	> 180	< 180
c/d ratio	$= \pi$	$< \pi$	$> \pi$
Curvature	Zero	Positive	Negative

Note: if Euclidean geometry is consistent, so are the others. Fifth postulate is **independent** of other four postulates.

The Question

We now face an embarrassment of riches: **multiple** logically consistent geometries.

?? *Which geometry describes physical space? And how could we possibly find out?*

Question for this week, concerns what geometry is about and how it relates to physics.

Epistemology of Geometry

Geometry as Knowledge: The Traditional View

Since the time of Plato's Academy geometry held a privileged epistemic status:

- A paradigm of **certain knowledge**: deductions from self-evident axioms, systematic structure clear; model for other domains to emulate
- Taken to describe the actual structure of physical space

Two questions that we should distinguish:

1. What is the *subject matter* of geometry?
2. How do we gain *knowledge* of geometric truths?

What is Geometry About?

Three possible answers:

1. Geometry **abstracts from bodies**: we discover geometric properties by studying physical objects and idealizing away imperfections
 2. Geometry describes **space itself**: geometric properties belong to space, and bodies inherit them by occupying spatial regions
 3. Geometry is a **formal system**: axioms are “free creations of the human mind,” devoid of physical content (until we apply them)
- ??** *On views (1) and (2), geometry has built-in empirical content. On view (3), the empirical content has to be added (... and how do we do that?)*

Geometry Entangled with Physics

A further problem lurks beneath these options:

- Geometry is *presupposed* by the laws of physics (e.g., “straight line,” “distance”)
- But we (apparently) only have *access* to geometric structure through physical measurements

Circular? Physics needs geometry, but geometry needs physics.

Intrinsic vs. Extrinsic Geometry

A useful distinction (that will be important later):

Extrinsic

- How a surface sits in a higher-dimensional space
- Requires a “view from outside”
- We can see that a sphere is curved by looking at it in 3D

Intrinsic

- Properties measurable *from within* the surface
- No reference to anything outside
- Distances, angles, c/d ratios—all measured by inhabitants

Intrinsic geometry more relevant for physics.

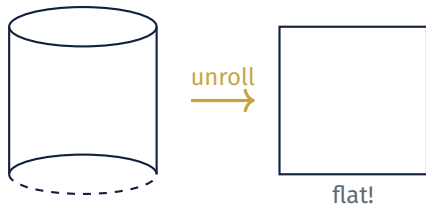
The Cylinder

A cylinder *looks* curved, but intrinsically it is flat:

- $\triangle$ angles sum to 180
- $c/d = \pi$ for circles
- Parallel lines stay parallel

Why? You can unroll it into a flat plane without stretching.

Lesson: contrast between extrinsic and intrinsic geometries; *measurements* needed to determine intrinsic geometry.



Gauss and the Mountains

- Gauss (c. 1820s) measured angles of a large triangle formed by three mountain peaks (sides $\sim 60\text{--}100$ km)
- Result: angles summed to 180 within measurement error

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- No: a non-Euclidean space with *very large* radius of curvature would give the same result within experimental error. Measurement alone cannot *conclusively* settle the question.
- Further reason for no: even if we *did* find angles > 180 , could we be sure this reflects the geometry of space, rather than some distortion of our instruments?

The Measurement Problem

To determine geometry by measurement, we need:

- **Measuring rods** to determine distances
- **Protractors** to measure angles
- **Light rays** to define “straight line”

Plus assumptions about each:

- Measuring rods doesn't change length during transport
- Light really travels in straight lines
- No “invisible” forces systematically distort everything

?? *How do we know these assumptions hold? If a “universal force” distorts all instruments identically, we could never detect it...*

Einstein: Two Kinds of Geometry

Einstein (1921) sharpened the problem:

Axiomatic Geometry

- Axioms as “free creations of the human mind”
- Purely formal, “void of all content of intuition or experience”
- Disconnected from the real world

Practical Geometry

- Abstract terms coordinated with physical objects
- “Straight line” $\leftrightarrow$ light ray
- “Distance” $\leftrightarrow$ rigid rod
- Makes empirical claims

“Geometry thus completed is evidently a natural science; we may in fact regard it as the most ancient branch of physics.”

— Einstein, “Geometry and Experience” (1921)

On this view:

- Pure (axiomatic) geometry is certain but says nothing about physical space
- Applied (practical) geometry says something about space but is *fallible*
- The price of empirical content is the loss of certainty

The Coordination Problem

To turn axiomatic geometry into practical geometry, we need **coordinating assumptions**:

- “Straight line” corresponds to the path of a light ray
- “Equal length” corresponds to what a rigid rod measures
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?? *But how do we know the rod is rigid? How do we know light travels in straight lines? These are physical assumptions—and they could be wrong.*

These are the questions that prompt Poincaré’s new position.

Four Possible Positions

Given multiple geometries and the coordination problem:

1. **A priori** (Kant): we know the geometry of space through the structure of our own minds—Euclidean geometry is a precondition for experience
2. **Empirical** (Gauss, Helmholtz): determined by measurements, other aspects of experience
3. **Entangled with physics**: geometry is bound up with physical theory; the question cannot be cleanly separated
4. **Conventionalist** (Poincaré): geometry is never tested in isolation, and the “choice” of geometry involves a convention

Break

Poincaré's Conventionalism

Poincaré: Neither A Priori Nor Empirical

Poincaré (1854–1912) rejected two extreme views:

- **Against a priori:** we *can* imagine experiences consistent with non-Euclidean geometry, so Euclidean geometry not determined *a priori*
- **Against empirical:** geometry is *exact*, but empirical science is inherently approximate; and geometry is never tested in isolation

His conclusion: the choice of geometry is (at least in part) **conventional**.

The Heated Disk: Setup

Huggett (Ch. 13) includes Poincaré's most famous thought experiment:

A disk-shaped world where temperature decreases from center to edge. All objects—including measuring rods—contract by a factor $\ell \cdot \frac{R^2 - r^2}{R^2}$ (where R is the radius of the disk, and r is the distance of the object from the center).

As an inhabitant walks toward the edge: measuring rods shrink, the edge recedes, ...

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The inhabitants carry out careful surveys of their world:

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The Heated Disk: What Inhabitants Measure

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- These deviations are larger for bigger figures, smaller for small ones near the center

?? *What should the inhabitants conclude? Is their space non-Euclidean?
Or is something happening to their measuring rods?*

Two Descriptions, Same Observations

The same measurements can be described in two equivalent ways:

Description 1

- Geometry: **hyperbolic**
- Rods: undistorted
- Space itself is curved
- $c/d > \pi$ because of geometry

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Both agree on *every possible measurement* the inhabitants could make.

Why Can't the Inhabitants Tell?

?? *Can the inhabitants perform some experiment to distinguish the two descriptions?*

No! Because:

- All objects shrink uniformly near the edge (rods, rulers, their own bodies)
- There is no “unshrunk” standard to compare against
- The contraction affects *everything*—it is a **universal** effect
- This is what makes it different from, say, thermal expansion of a brass rod (which can be detected by comparing with a steel rod)

The Conventionalist Argument

Schematically:

Observed result = Geometry G + Physical theory P

- Observations constrain the *combination* $G + P$
- We can always trade: change G , adjust P to compensate
- So no observation can determine G *alone*

The choice of G is therefore a **convention**—guided by simplicity and convenience, not by “truth.”

The $G + P$ Trade-Off in Detail

For the heated disk:		Description 1	Description 2
	Geometry G	Hyperbolic	Euclidean
	Physics P	Rods are rigid	Rods shrink: $\ell \cdot \frac{R^2-r^2}{R^2}$
	Observed c/d	$> \pi$	$> \pi$
	Observed $\triangle$	< 180	< 180

Same observations, different $G + P$ packages

Which is “right”? Poincaré: neither is more true; the question is which is more *convenient*.

Is This Trivial?

A natural objection: isn't this just like choosing meters vs. yards?

Poincaré thinks the point goes deeper:

- Different choices of G lead to *different theoretical structures*—not just relabeling
- But the conventional choice is constrained: some geometries would be wildly inconvenient
- Euclidean geometry will typically be the *simplest* choice

?? *Poincaré does not say geometry is arbitrary. It is a real choice, guided by pragmatic considerations, but not a factual one.*

- Conventionalism **specific to geometry**—Poincaré does not extend it to all of science
- Geometry presupposed by physics: physical laws need background geometric assumptions
- Poincaré retains Kantian view about other areas (e.g., arithmetic)
- But parts from Kant on geometry, due to discovery that non-Euclidean geometries are consistent

Against Empiricism

As Huggett discusses, Poincaré gives two main reasons geometry cannot be simply empirical:

1. Geometry is *exact*; empirical science is approximate. If geometry were empirical, Poincaré asks, “what a rough approximation it would be!”
2. Geometry provides the *framework* within which physical theories are formulated—it cannot be tested in isolation

“Can we maintain that certain phenomena which are possible in Euclidean space would be impossible in non-Euclidean space? ... [E]xperience cannot decide between Euclid and Lobachevsky.”

— Poincaré, *Science and Hypothesis*, in Huggett Ch. 13

Poincaré also rejects Kant. As discussed in Huggett Ch. 13:

“There is nothing to prevent us from imagining a series of representations, similar in every way to our ordinary representations, but succeeding one another according to laws which differ from those to which we are accustomed.”

— Poincaré, *Science and Hypothesis*, in Huggett Ch. 13

- We can imagine experiences consistent with non-Euclidean geometry
- A being raised in a non-Euclidean world would form a non-Euclidean concept of space
- Nothing in our cognitive structure forces Euclidean geometry upon us

Poincaré's "Genetic" Account

So where does our concept of space come from?

"None of our sensations, if isolated, could have brought us to the concept of space; we are brought to it solely by studying the laws by which those sensations succeed one another."

— Poincaré, *Science and Hypothesis*, in Huggett Ch. 13

- Our concept of space is *extracted from* patterns of experience—especially the experience of **rigid body motions**
- Some changes can be “compensated” by moving our body; these compensable displacements form a mathematical group
- The properties of that group are what we call “geometry”—but they do not uniquely single out Euclidean geometry

Taking Stock

Position	Geometry is...	Challenge
Kantian	Synthetic a priori	Non-Euclidean geometries are consistent
Empiricist	Determined by measurement	Never tested in isolation; needs co-ordinating assumptions
Conventionalist	Conventional choice	How to avoid triviality? What are its limits?

Open Questions

- ?? *Is the conventionalist right that no observation can determine the geometry alone? Or are there physical considerations that break the symmetry between alternative descriptions?*
- ?? *If the “conventional” choice is always guided by simplicity, is it really a convention—or is simplicity itself a guide to truth?*

On Thursday we will look at Reichenbach's attempt to make the “conventional element” precise, and give further assessment of this view.

For Next Time

- **Read:** Huggett, Ch. 13 (C 3); Norton, “Philosophy of Space and Time” (C 2), section: “Conventionality of Geometry.”

Thursday: **Reichenbach and Coordinative Definitions**

- Making the “conventional element” precise
- Universal vs. differential forces
- The “relativity of geometry”

Questions?

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